Chapter 12

The Data Warehouse

Database Systems:
Design, Implementation, and Management, Sixth Edition, Rob and Coronel

Business Problems & Data Analysis Needs

- Operational databases served as the source of information to facilitate the decision making process
  - Decision support systems (DSS) were developed around this data
- Information requirements have grown quite complex over time and it is difficult to extract the all the needed information from a database

Business Problems & Data Analysis Needs

- The data warehouse provides a more comprehensive data pool by including not only operational data but data from external sources as well
- The data warehouse also stores the data in structures that simplify information generation as well make it possible to generate a type and extent of data not otherwise available

Solving Business Problems and Adding Value with Data Warehouse-Based Solutions

TABLE 12.1 Solving Business Problems and Adding Value with data Warehouse-Based Solutions

COMPANY	PROBLEM	BENEFIT
MOEN Manufacturer of bathroom and kitchen fixtures and supplies Source: Cognos Corp. *www.cognos.com*	• Information generation very limited and time-consuming. • Only five people knew how to extract data using a 3GL. • Response time unacceptable for Managers' decision-making purposes	• Provided quick answers to ad hoc questions for decision making. • Provided access to data for decision-making purposes. • Received in-depth view of product performance and customer margins.
Pacific Gas Transmission Co. Natural gas provider in Pacific Northwest Source: Oracle Corp. *www.oracle.com*	• Rapid changes in markets due to deregulation. • Diminishing profits in traditional services.	• Managers able to analyze data quickly. • Positioned company to quickly identify market trends. • Created new services and pricing structures.
SEGA Interactive entertainment systems and video games Source: Oracle Corp. *www.oracle.com*	• Needed way to rapidly analyze great amount of data. • Needed to track advertising, coupons, and rebates associated with effects of pricing changes. • Formerly used Excel spreadsheets, leading to human errors.	• Eliminated data-entry errors. • Identified successful marketing strategies to dominate interactive entertainment niches. • Used product analysis to identify better markets/product offerings.

Solving Business Problems and Adding Value with Data Warehouse-Based Solutions

TABLE 12.1 SOLVING BUSINESS PROBLEMS AND ADDING VALUE WITH DATA WAREHOUSE-BASED SOLUTIONS (CONTINUED)

COMPANY	PROBLEM	BENEFIT
Owens and Minor, Inc. Medical and surgical supply distributor Source: CFO Magazine *www.cfomagazine.com*	• Lost its largest customer, which represented 10% of its annual revenue ($360 million). • Stock plunged 23%. • Cumbersome process to access information from antiquated mainframe system.	• In just five months increased earnings per share. • Gained more business because the data warehouse was opened to its clients. • Managers gained quick access to data for decision-making purposes.
LA Cellular Cellular telephone company in Los Angeles area Source: *PC Week Online* *www.zdnet.com/pcweek/stories*	• Needed reduced response time to business questions. • Needed to identify which promotions were working, and which were not. • Needed to identify which customers to call—out of a database containing millions of customers—in order to offer new promotions.	• Achieved a 20% increase in subscribers, as a result of properly matching customers with promotions. • Could identify which promotions were effective. • Response times were cut from 14 minutes to 1 minute.

Decision Support Systems

- Methodology (or series of methodologies) designed to extract information from data and to use such information as a basis for decision making
- Decision support system (DSS):
  - Arrangement of computerized tools used to assist managerial decision making within a business
  - Usually requires extensive data “massaging” to produce information
  - Used at all levels within an organization
  - Often tailored to focus on specific business areas
  - Provides ad hoc query tools to retrieve data and to display data in different formats

Decision Support Systems

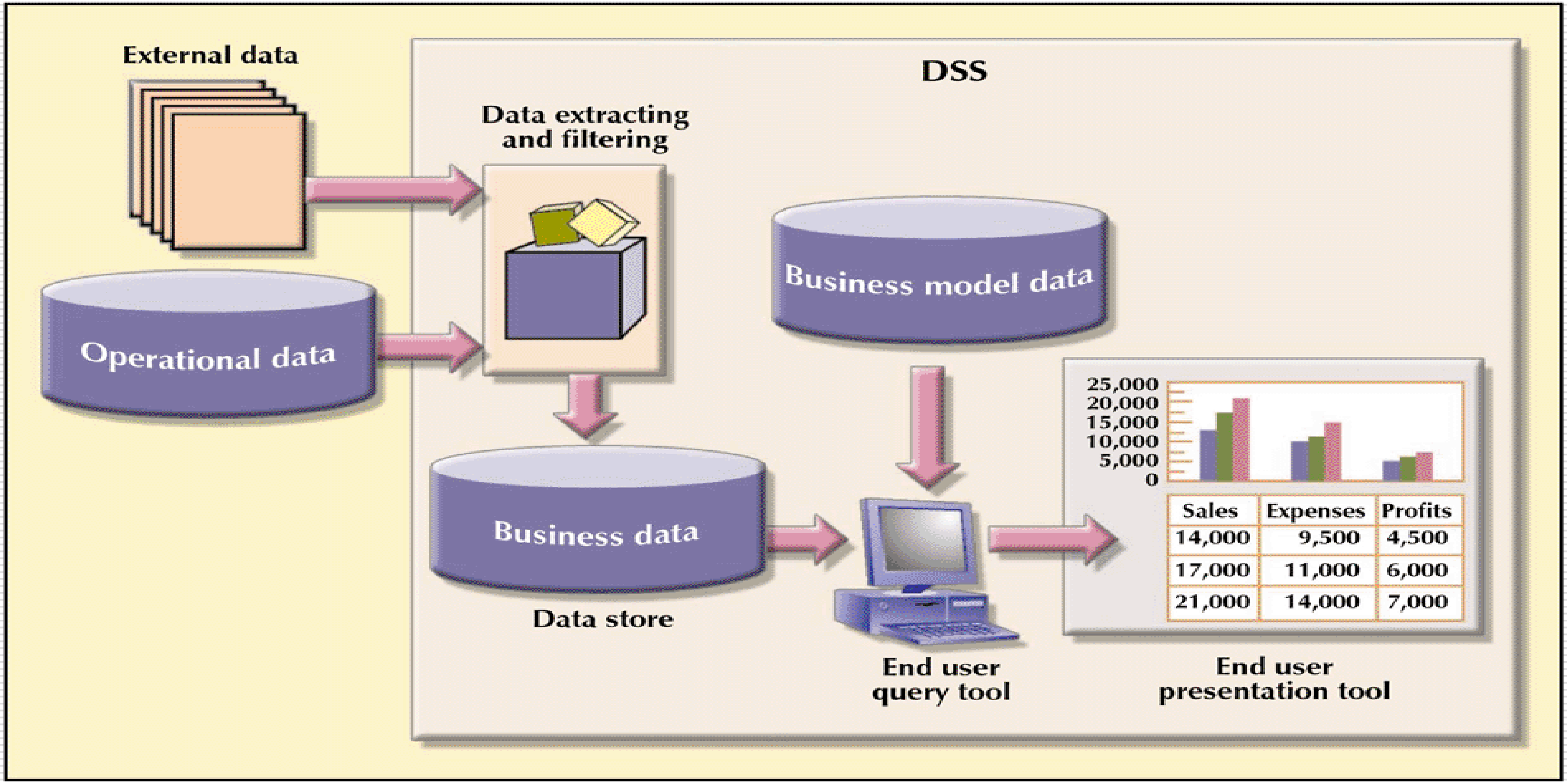
- Combines historical operational data with business models that reflect the business activities
  - Compare the relative rates of productivity growth by company division over some specified period of time
  - Define the relationship between advertising types and sales levels
  - Define relative market shares by selected product lines

Decision Support Systems

- Composed of four main components:
  - Data store component
    - Basically a DSS database containing business data and data model data coming from internal and external sources.
    - Data is summarized and arranged in structures that are optimized for data analysis and query speed
  - Data extraction and filtering component
    - Used to extract and validate data taken from operational database and external data sources
  - End-user query tool
    - Used to create queries that access database
  - End-user presentation tool
    - Used to organize and present data

Main Components of a Decision Support System (DSS)

FIGURE 12.1 MAIN COMPONENTS OF A DECISION SUPPORT SYSTEM (DSS)



Transforming Operational Data Into Decision Support Data

FIGURE 12.2 TRANSFORMING OPERATIONAL DATA INTO DECISION SUPPORT DATA

Operational Data

	A	B	C	D	E
3	Year	Region	Agent	Product	Value
4	2002	East	Carlos	Erasers	50
5	2002	East	Tere	Erasers	12
6	2002	North	Carlos	Widgets	120
7	2002	North	Tere	Widgets	100
8	2002	North	Carlos	Widgets	30
9	2002	South	Victor	Balls	145
10	2002	South	Victor	Balls	34
11	2002	South	Victor	Balls	80
12	2002	West	Mary	Pencils	89
13	2002	West	Mary	Pencils	56
14	2003	East	Carlos	Pencils	45
15	2003	East	Victor	Balls	55
16	2003	North	Mary	Pencils	60
17	2003	North	Victor	Erasers	20
18	2003	South	Carlos	Widgets	30
19	2003	South	Mary	Widgets	75
20	2003	South	Mary	Widgets	50
21	2003	South	Tere	Balls	70
22	2003	South	Tere	Erasers	90
23	2003	West	Carlos	Widgets	25
24	2003	West	Tere	Balls	100

Operational data have a narrow time span, low granularity, and single focus. Such data are usually presented in tabular format, in which each row represents a single transaction. This format often makes it difficult to derive useful information.

Time

Product

Agent

Region

Decision Support Data

	A	B	C	D	E	F
1	Year	2003				
2						
3	Sum of Value	Region				
4	Product	East	North	South	West	Total
5	Balls	55		[illegible]0	100	225
6	Erasers		20	90		110
7	Pencils	45	60			105
8	Widgets			155	25	180
9	Total	100	80	315	125	620
10						
11						
12	Year	(All)				
13	Product	(All)				
14						
15	Sum of Value	Region				
16	Agent	East	North	South	West	Total
17	Carlos	95	150	[illegible]0	25	300
18	Mary		60	[illegible]25	145	330
19	Tere	12	100	160	100	372
20	Victor	55	20	259		334
21	Total	162	330	574	270	1,336

Sales

Decision support system (DSS) data focus on a broader time span, tend to have high levels of granularity, and can be examined in multiple dimensions. For example, note these possible aggregations:

- Sales by product, region, agent, etc.
- Sales for all years or only a few selected years.
- Sales for all products or only a few selected products.

Contrasting Operational and DSS Data Characteristics

TABLE 12.2 CONTRASTING OPERATIONAL AND DSS DATA CHARACTERISTICS

CHARACTERISTIC	OPERATIONAL DATA	DSS DATA
Data currency	Current operations Real-time data	Historic data Snapshot of company data Time component (week/month/year)
Granularity	Atomic-detailed data	Summarized data
Summarization level	Low; some aggregate yields	High; many aggregation levels
Data model	Highly normalized Mostly relational DBMS	Nonnormalized Complex structures Some relational, but mostly multidimensional DBMS
Transaction type	Mostly updates	Mostly query
Transaction volumes	High update volumes	Periodic loads and summary calculations
Transaction speed	Updates are critical	Retrievals are critical
Query activity	Low to medium	High
Query scope	Narrow range	Broad range
Query complexity	Simple to medium	Very complex
Data volumes	Hundreds of megabytes and up to gigabytes	Hundreds of gigabytes to terabytes

DSS Database Requirements

- Database schema
  - Must support complex (non-normalized) data representations
  - Database must contain data that are aggregated and summarized and maintain relations with many other data elements
  - Queries must be able to extract multidimensional time slices

Ten-Year Sales History for a Single Department, in Millions of Dollars

TABLE 12.3 Ten-Year Sales History for a Single Department, in Millions of Dollars

YEAR	SALES
1994	8,227
1995	9,109
1996	10,104
1997	11,553
1998	10,018
1999	11,875
2000	12,699
2001	14,875
2002	16,301
2003	19,986

- 10 year sales history for a single store containing a single department
  - Data are fully normalized within the single table
- Next slide shows yearly summaries of sales for two stores, each with two departments
  - As number of years, stores and departments increase, redundancies increase and table may become non-normalized to speed up queries

Yearly Sales Summaries, Two Stores and Two Departments per Store, In Millions of Dollars

TABLE 12.4 Yearly Sales Summaries, Two Stores and Two Departments per Store, in Millions of Dollars

YEAR	STORE	DEPARTMENT	SALES
1994	A	1	1,985
1994	A	2	2,401
1994	B	1	1,879
1994	B	2	1,962
...	...	...	...
1998	A	1	3,912
1998	A	2	4,158
1998	B	1	3,426
1998	B	2	1,203
...	...	...	...
2003	A	1	7,683
2003	A	2	6,912
2003	B	1	3,768
2003	B	2	1,623

DSS Database Requirements

- Data extraction and loading
  - DSS database is created largely by extracting data form the operational database and by importing additional data from external sources
  - Thus, the DBMS must support advanced data extraction and filtering tools (batch and scheduled)
    - Support different data sources – flat files, relational, multiple vendors
    - Check for inconsistent data or data validation rules

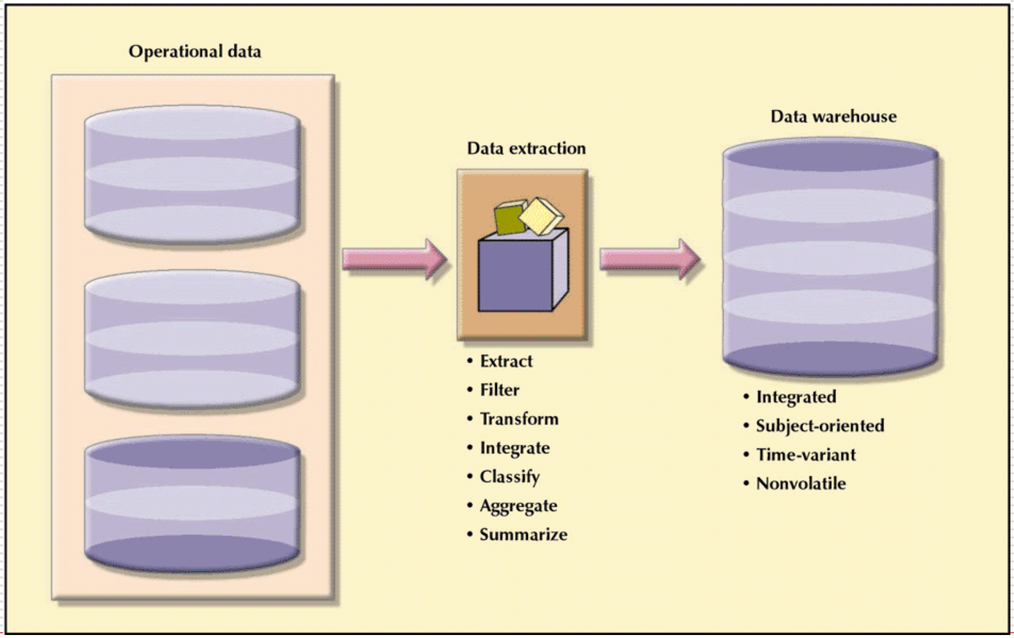
DSS Database Requirements

- End-user analytical interface
  - DSS DBMS must support advanced data modeling and data presentation tools
    - This makes it easier to define the business problem
    - Once information retrieved, data analysis tools can be used to evaluate the query results
- Database size
  - Must be capable of supporting very large databases (VLDBs)
    - Employ multiple disk arrays and multi-processor technologies such as symmetric MP or massively parallel processors

The Data Warehouse

- Bill Inmon, father of the data warehouse, defines it as an *integrated, subject-oriented, time-variant, nonvolatile* database that provides support for decision making
- Usually a read-only database optimized for data analysis and query processing

FIGURE 12.3 CREATING A DATA WAREHOUSE



A Comparison of Data Warehouse and Operational Database Characteristics

TABLE 12.5 A Comparison of Data Warehouse and Operational Database Characteristics

CHARACTERISTIC	OPERATIONAL DATABASE DATA	DATA WAREHOUSE DATA
Integrated	Similar data can have different representations or meanings. For example, Social Security numbers may be stored as ###-##-#### or as #########, and a given condition may be labeled as T/F or 0/1 or Y/N. A sales value may be shown in thousands or in millions.	Provide a unified view of all data elements with a common definition and representation for all business units.
Subject-oriented	Data are stored with a functional, or process, orientation. For example, data may be stored for invoices, payments, credit amounts, and so on.	Data are stored with a subject orientation that facilitates multiple views of the data and facilitates decision making. For example, sales may be recorded by product, by division, by manager, or by region.
Time-variant	Data are recorded as current transactions. For example, the sales data may be the sale of a product on a given date, such as $342.78 on 12-MAY-2004.	Data are recorded with a historical perspective in mind. Therefore, a time dimension is added to facilitate data analysis and various time comparisons.
Nonvolatile	Data updates are frequent and common. For example, an inventory amount changes with each sale. Therefore, the data environment is fluid.	Data cannot be changed. Data are only added periodically from historical systems. Once the data are properly stored, no changes are allowed. Therefore, the data environment is relatively static.

The Data Mart

- Because of the time, money and considerable mangerial effort required to create a data warehouse, many companies begin on a smaller scale with a *data mart*
- A data mart is a small, single-subject data warehouse subset that provides decision support to a small group of people
  - Lower cost, shorter implementation time
  - Data marts can be customized to small groups in ways a centralized data warehouse can not
  - Company culture may be to go slow with big changes, data mart is not as threatening
  - Benefits can be determined based on experience which can give a justification to expand its use

Summary of DSS Architectural Styles

TABLE 12.6 **DSS Architectural Styles**

SYSTEM TYPE	SOURCE DATA	DATA EXTRACTION/ INTEGRATION PROCESS	DSS DATA STORE	END-USER QUERY TOOL	END-USER PRESENTATION TOOL
Traditional mainframe-based OLTP	Operational data	None Reports, reads, and summarizes data directly from operational data	None Temporary files used for reporting purposes	Very basic Predefined reporting formats. Basic sorting, totaling, and averaging	Very basic Menu-driven, predefined reports, text and numbers only
Managerial information system (MIS) with third-generation language (3GL)	Operational data	Basic extraction and aggregation Reads, filters, and summarizes operational data into intermediate data store	Lightly aggregated data in RDBMS	Same as above, plus some ad hoc reporting using SQL	Same as above, plus some ad hoc columnar report definitions
First-generation departmental DSS	Operational data	Data extraction and integration process to populate a DSS data store; run periodically	1st DSS database generation Usually RDBMS	Query tool with some analytical capabilities and predefined reports	Advanced presentation tools with plotting and graphics capabilities
First-generation enterprise data warehouse using RDBMS	Operational data External data (census data)	Advanced data extraction and integration tools Features include access to diverse data sources, transformations, filters, aggregations, classifications, scheduling, and conflict resolution	Data warehouse integrated DSS database to support the entire organization Uses RDBMS technology optimized for query purposes Star schema model	Same as above, plus support for more advanced queries and analytical functions with extensions	Same as above, plus additional multidimensional presentation tools with drill-down capabilities
Second-generation data warehouse using MDBMS	Operational data External data (industry group data)	Same as first generation enterprise data warehouse using RDBMS	Data warehouse stores data using multidimensional database (MDBMS) technology based on data structures; referred to as "cubes" with multiple dimensions	Same as above, but uses different query interface to access MDBMS (proprietary)	Same as above, but uses "cubes" and multidimensional matrixes Limited in terms of cube size

12 Rules of a Data Warehouse

Inmon, Bill and Kelley, Chuck, "The Twelve Rules of Data Warehouse for a Client/Server World", *Data Management Review*, 4(5), May 1994, pp 6-16.

- Data Warehouse and Operational Environments are Separated
- Data is integrated
- Contains historical data over a long period of time
- Data is a snapshot data captured at a given point in time
- Data is subject-oriented

12 Rules of Data Warehouse

- Mainly read-only with periodic batch updates
- Development Life Cycle has a data driven approach versus the traditional process-driven approach
- Data contains several levels of detail
  - Current, Old, Lightly Summarized, Highly Summarized

Online Analytical Processing

- Advanced data analysis environment that supports decision making, business modeling, and operations research
- OLAP systems share four main characteristics:
  - Use multidimensional data analysis techniques
  - Provide advanced database support
  - Provide easy-to-use end-user interfaces
  - Support client/server architecture

Operational vs. Multidimensional View of Sales

FIGURE 12.4 OPERATIONAL VS. MULTIDIMENSIONAL VIEW OF SALES

Database name: Ch12_Text

Table name: DW_INVOICE

INV_NUM	INV_DATE	CUS_NAME	INV_TOTAL
2034	15-May-04	Dartonik	$1,400.00
2035	15-May-04	Summer Lake	$1,200.00
2036	16-May-04	Dartonik	$1,350.00
2037	16-May-04	Summer lake	$3,100.00
2038	16-May-04	Trydon	$400.00

Table name: DW_LINE

INV_NUM	LINE_NUM	PROD_DESCRIPTION	LINE_PRICE	LINE_QUANTITY	LINE_AMOUNT
2034	1	Optical Mouse	$45.00	20	$900.00
2034	2	Wireless RF remote and laser pointer	$50.00	10	$500.00
2035	1	Everlast Hard Drive, 60 GB	$200.00	6	$1,200.00
2036	1	Optical Mouse	$45.00	30	$1,350.00
2037	1	Optical Mouse	$45.00	10	$450.00
2037	2	Roadster 56KB Ext. Modem	$120.00	5	$600.00
2037	3	Everlast Hard Drive, 60 GB	$205.00	10	$2,050.00
2038	1	NoTech Speaker Set	$50.00	8	$400.00

Multidimensional View of Sales

	Time Dimension		
Customer Dimension	**15-May-04**	**16-May-04**	**Totals**
Dartonik	$1,400.00	$1,350.00	$2,750.00
Summer Lake	$1,800.00	$3,100.00	$4,900.00
Trydon		$400.00	$400.00
Totals	$3,200.00	$4,850.00	$8,050.00

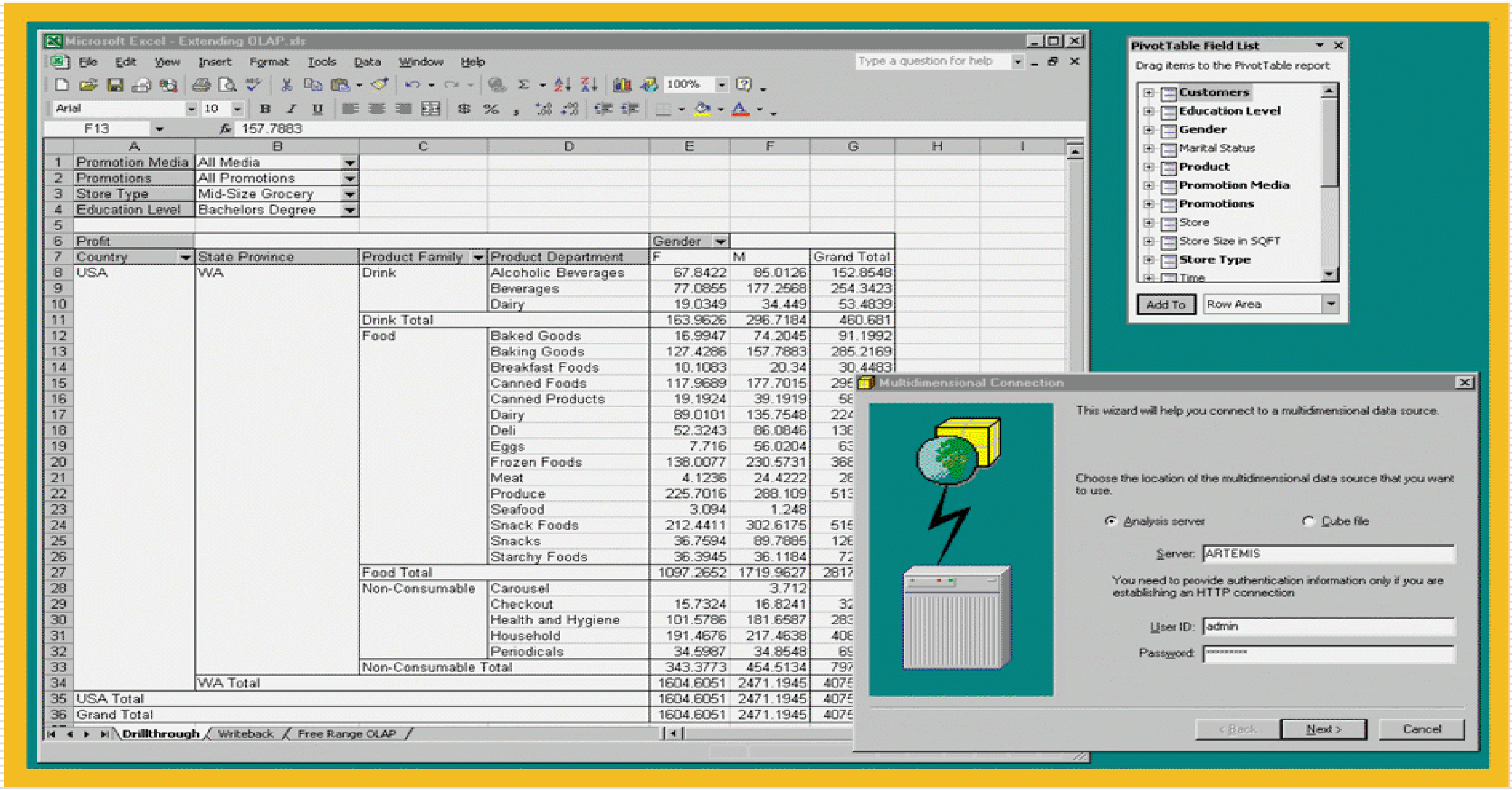
Sales are located in the intersection of a customer row and time column

Aggregations are provided for both dimensions

View business data as data that are related to other business data e.g., sales data as related to customers and time

Integration of OLAP with a Spreadsheet Program

FIGURE 12.5 Integration of OLAP with a Spreadsheet Program



Most OLAP vendors have closely integrated their systems with desktop spreadsheets to take advantage of the analysis and presentation functionality of the spreadsheets that users are already familiar with

Advanced Database Support

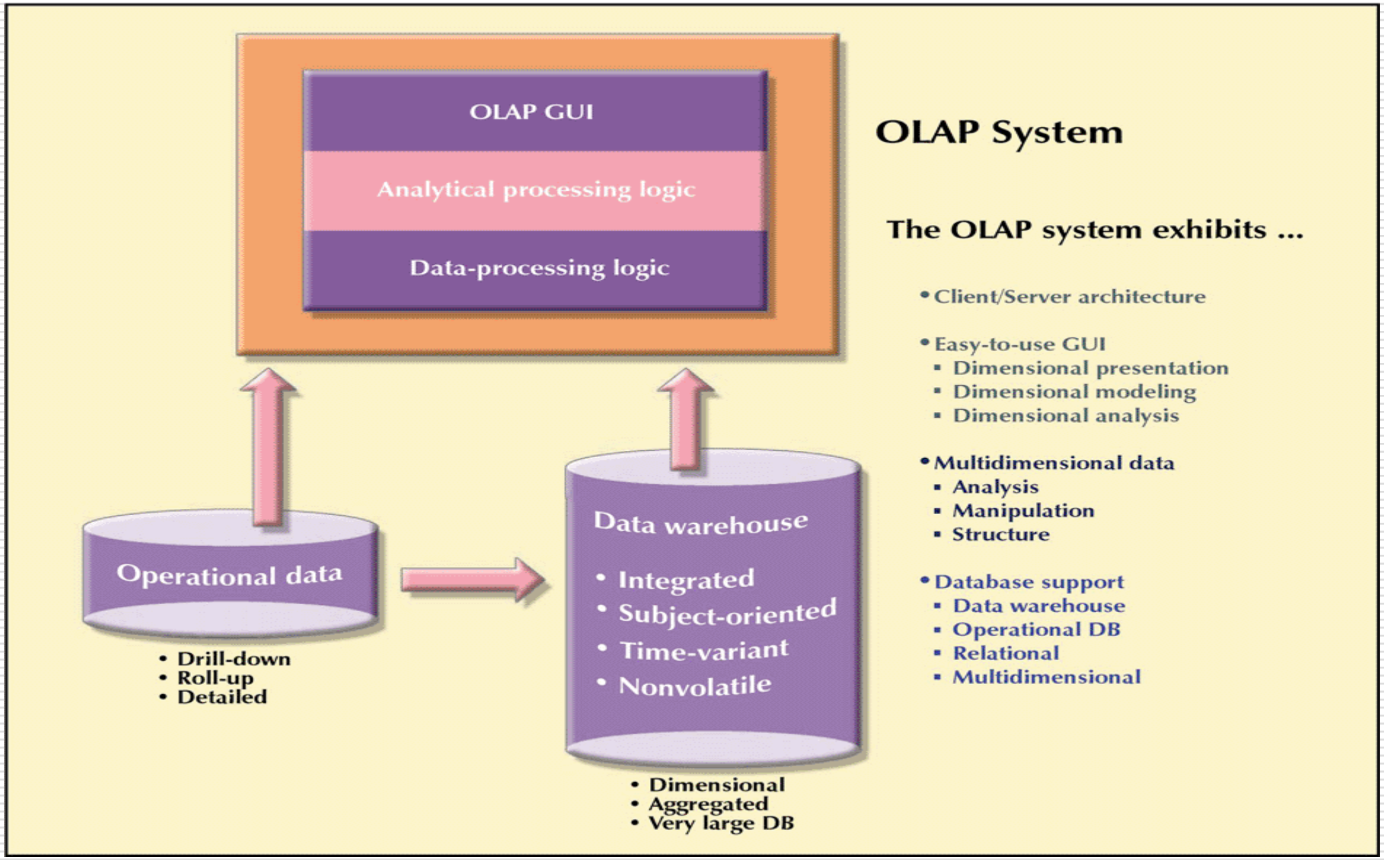
- To deliver efficient decision support, OLAP tools must have advanced data access features such as:
  - Access to many different kinds of DBMSs, flat files, internal and external data sources
  - Access to aggregated data warehouse data as well as detail data found in operational databases
  - Advanced data navigation – drill-down and roll-up
  - Support for VLDBs
  - Rapid and consistent query response times

Client-Server Architecture

- The C/S environment enables us to divide an OLAP system into several components that define its architecture
  - These components can then be placed on the same computer or distributed among several computers
- OLAP modules
  - GUI
  - Analytical processing logic
  - Data processing logic

OLAP Client/Server Architecture

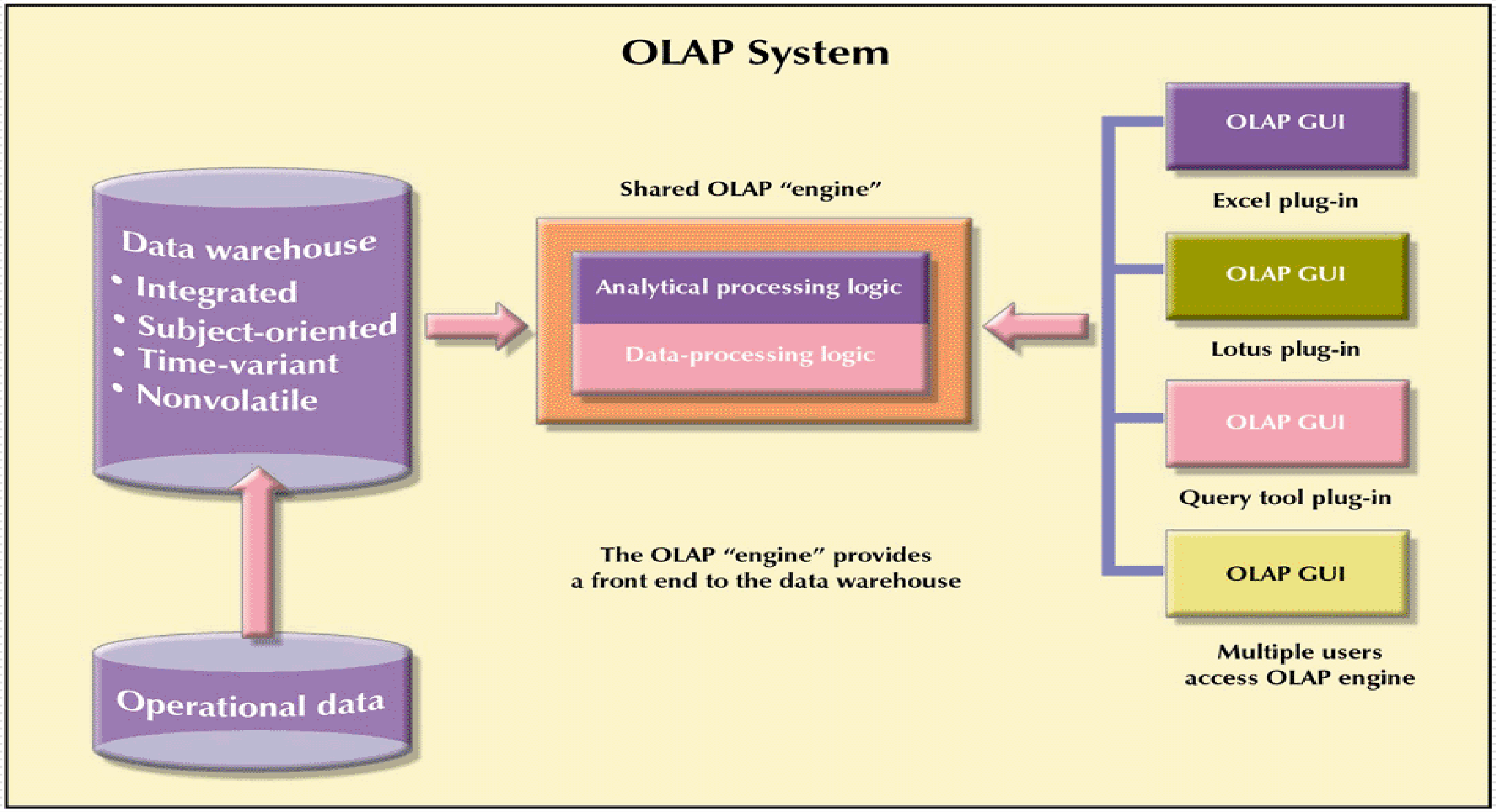
FIGURE 12.6 OLAP CLIENT/SERVER ARCHITECTURE



In this scenario, each analyst requires a powerful computer to store the OLAP system and perform all data processing locally. Also, each analyst uses a separate copy of the data – *islands of information* problem

OLAP Server Arrangement

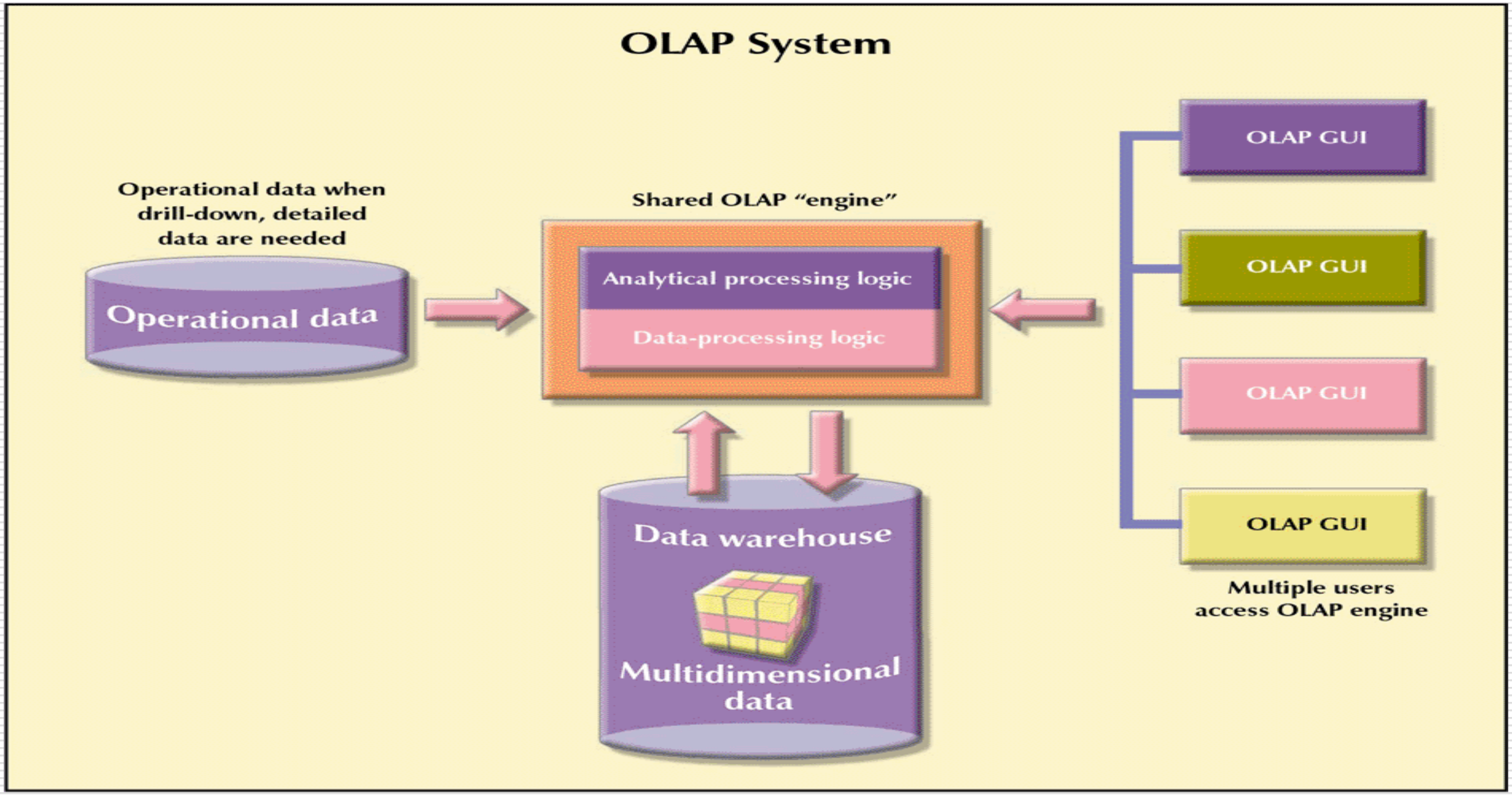
FIGURE 12.7 OLAP SERVER ARRANGEMENT



Here, the OLAP GUI runs on client workstations while the OLAP engine runs on a shared computer. The engine serves as the front-end to the data warehouse's decision support data. OLAP and the data warehouse are independent pieces of the system.

OLAP Server with Multidimensional Data Store Arrangement

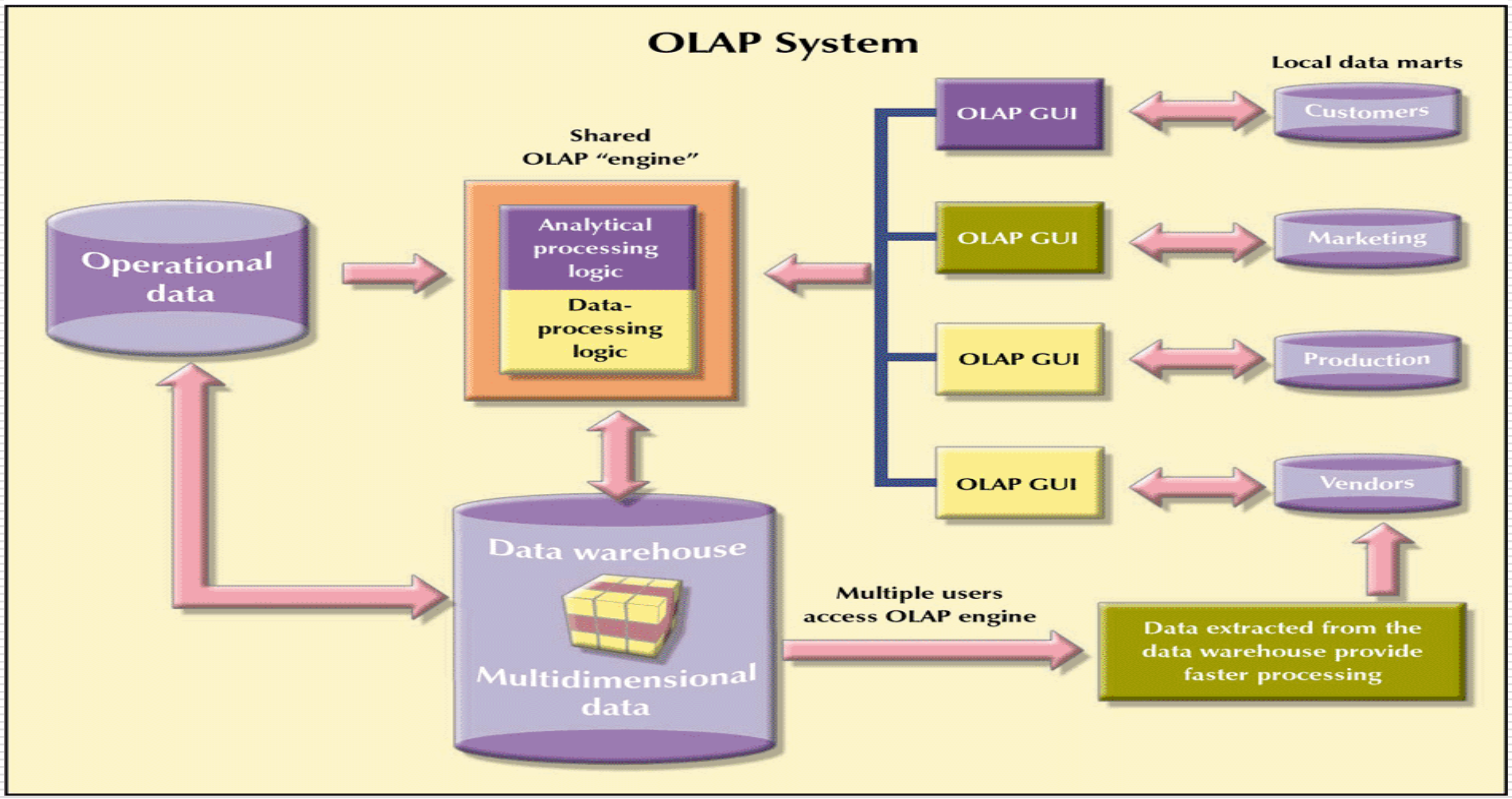
FIGURE 12.8 OLAP SERVER WITH MULTIDIMENSIONAL DATA STORE ARRANGEMENT



In most implementations, the data warehouse and OLAP are interrelated and complementary environments. Here, the OLAP engine extracts data from the operational db and stores it in a multidimensional structure for further analysis

OLAP Server With Local Mini Data Marts

FIGURE 12.9 OLAP SERVER WITH LOCAL MINI DATA MARTS



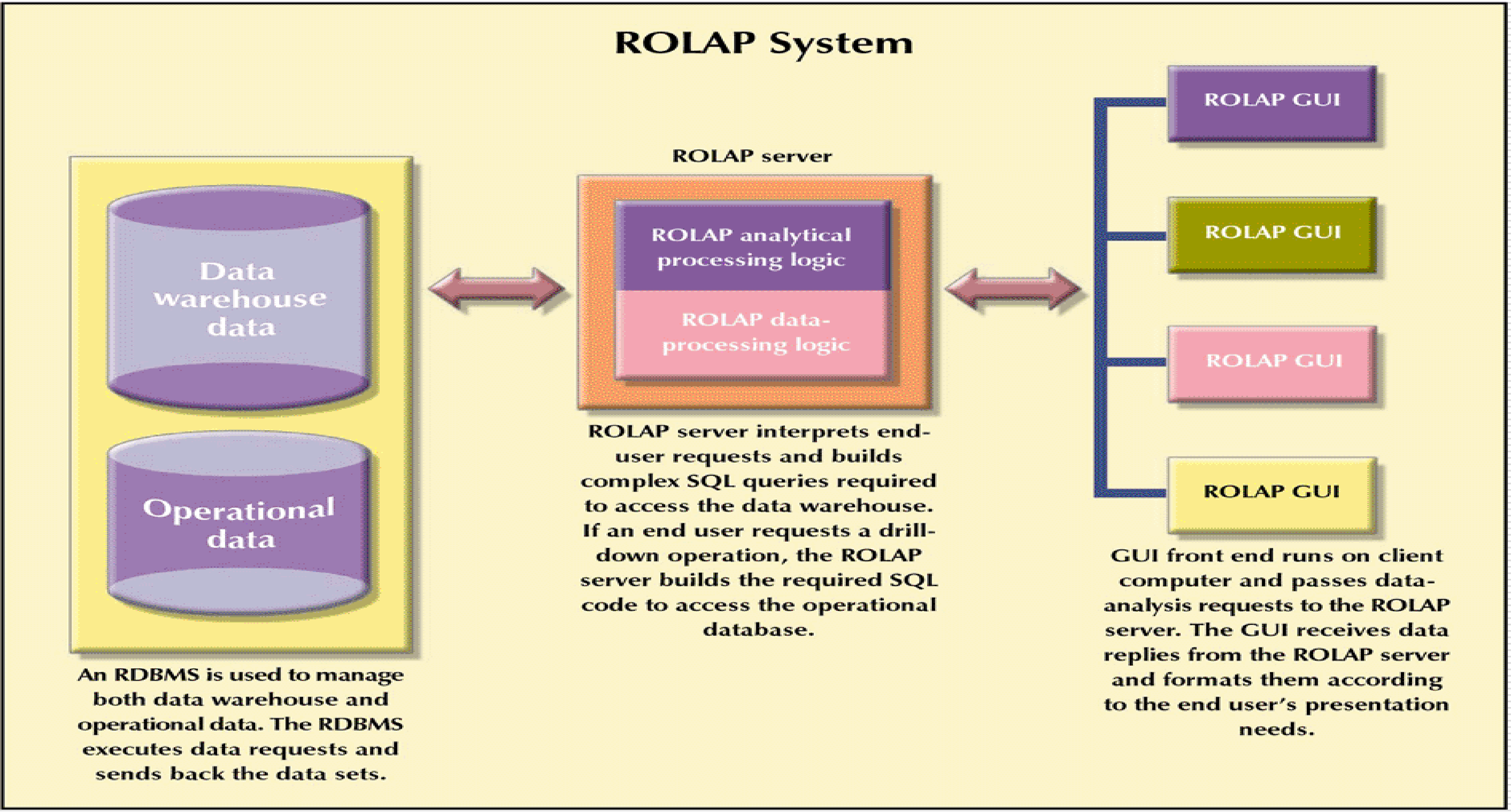
To provide better performance, some OLAP systems store small extracts of the data warehouse at end-user workstations. This increases the speed of data access and data visualization. Assumes that most end-users work with fairly small, stable data warehouse data subsets.

Relational OLAP

- Builds on existing relational technologies
- Adds the following extensions to RDBMS
  - Multidimensional data schema support within the RDBMS
    - Star schema to enable RDMS (normalized data) to support multidimensional data representations (nonnormalized, aggregated and duplicated)
  - Data access language and query performance are optimized for multidimensional data
    - ROLAP extends SQL so that it can differentiate between access requirements for data warehouse data and operational data
  - Support for VLDBs

Typical ROLAP Client/Server Architecture

FIGURE 12.10 Typical ROLAP Client/Server Architecture

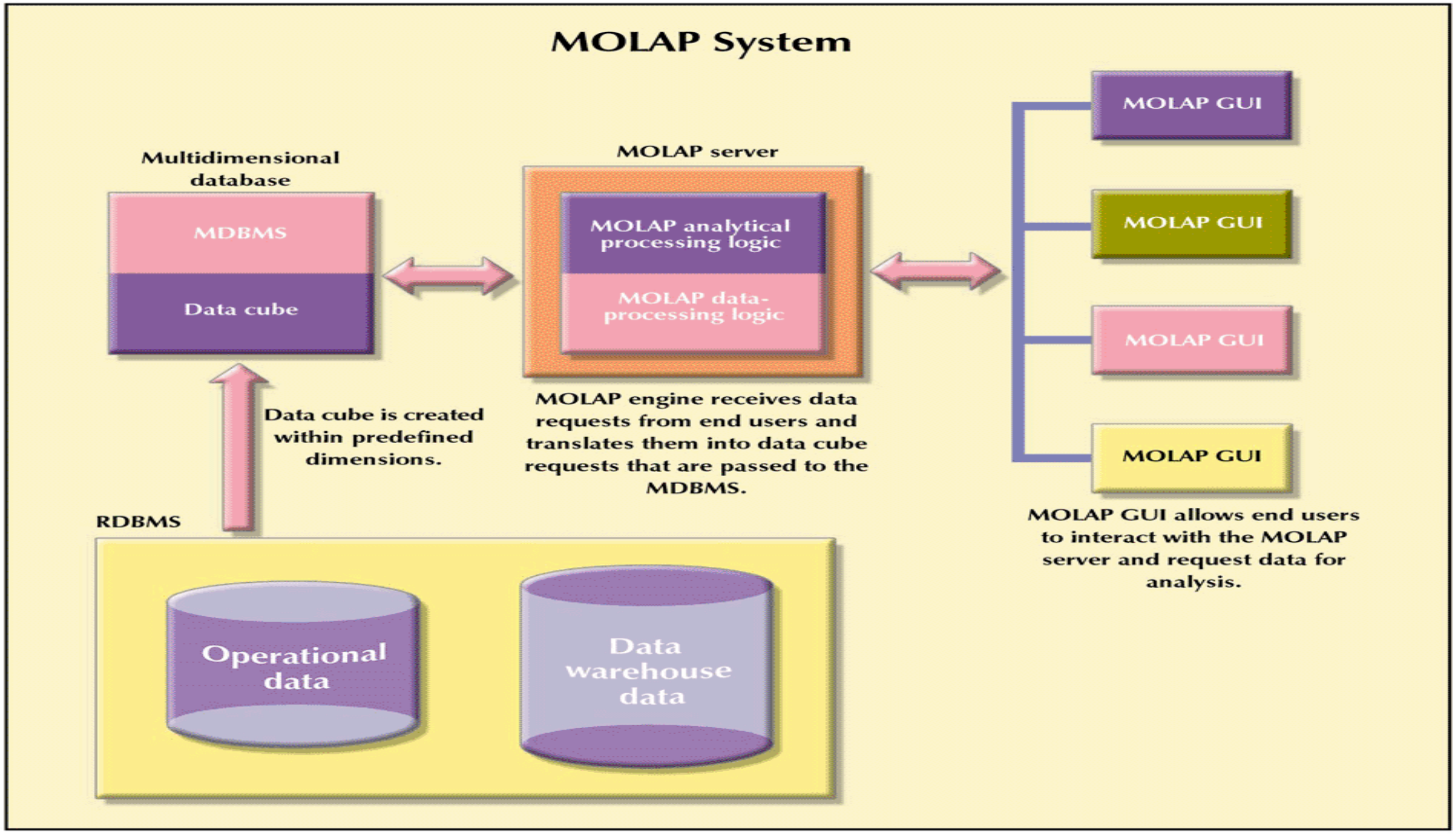


Multidimensional OLAP

- MOLAP extends OLAP functionality to multidimensional DBMSs (MDBMSs)
  - An MDBMS stores data in matrix-like n-dimensional arrays
  - MDBMS end users visualize the stored data as a three-dimensional cube known as a **data cube**
    - They data cubes can grow to n-dimensions becoming hypercubes
    - Data cubes are created by extracting data from the operational databases or the data warehouse
    - They are pre-created and static and queried based on their dimensions e.g., product, location and time for a cube for sales
    - To speed data access they are held in memory – **cube cache**

MOLAP Client/Server Architecture

FIGURE 12.11 MOLAP CLIENT/SERVER ARCHITECTURE



MDBMS

- Because a cube is pre-defined, the addition of a new dimension requires that the entire data cube be recreated – a time consuming process
  - If this needs to be done often, the MDBMS loses some of its speed advantage over the RDBMS
  - MDBMS is best suited for small and medium data sets
  - Scalability is limited due to the restrictions on the size of the data cube to avoid lengthy data access times caused by having less memory available for the OS and application programs
  - Employ proprietary data storage techniques that require proprietary data access methods using a multidimensional query language
  - Most handle sparsity of the data cubes effectively to reduce processing overhead and resource requirements

Relational vs. Multidimensional OLAP

TABLE 12.8 RELATIONAL VS. MULTIDIMENSIONAL OLAP

CHARACTERISTIC	ROLAP	MOLAP
Schema	Uses star schema Additional dimensions can be added dynamically	Uses data cubes Additional dimensions require re-creation of the data cube
Database size	Medium to large	Small to medium
Architecture	Client/server Standards-based Open	Client/server Proprietary
Access	Supports ad hoc requests Unlimited dimensions	Limited to predefined dimensions
Resources	High	Very high
Flexibility	High	Low
Scalability	High	Low
Speed	Good with small data sets; average for medium to large data sets	Faster for small to medium data sets; average for large data sets

Star Schemas

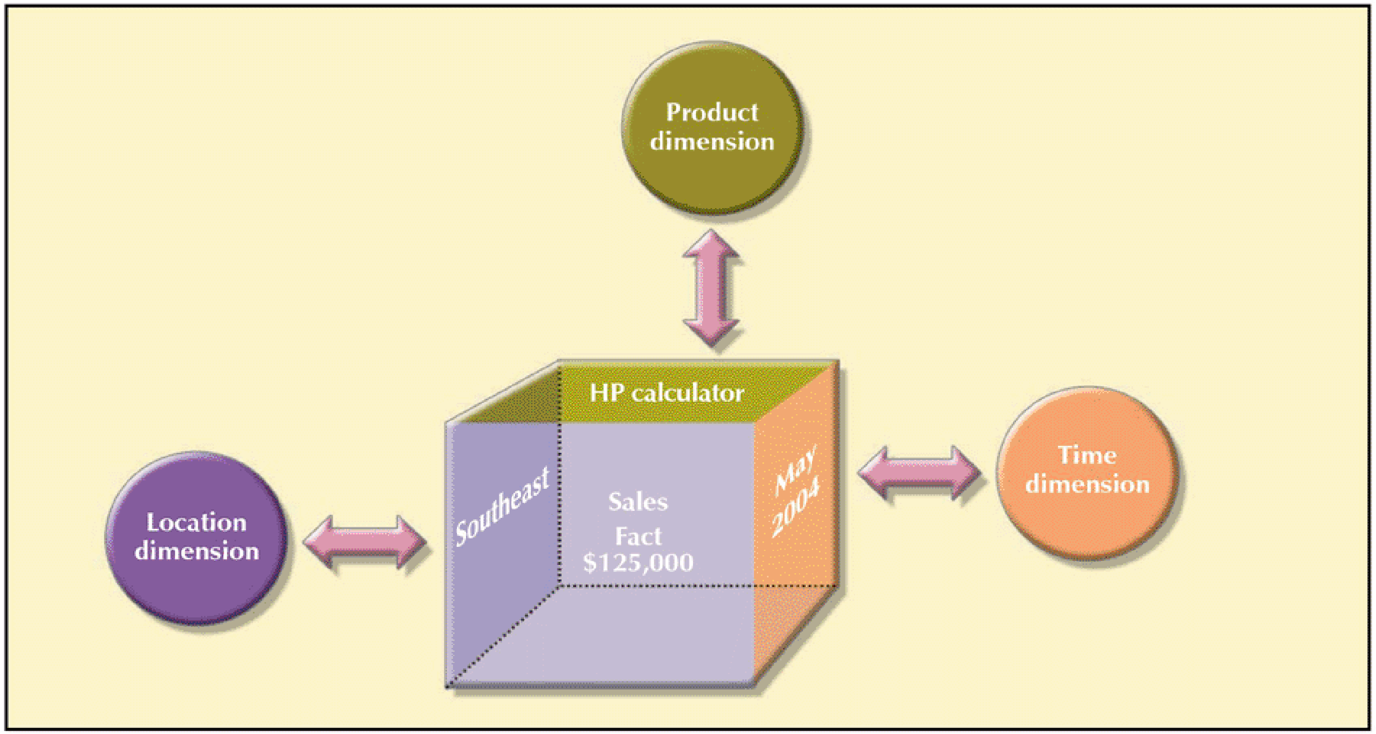
- Data modeling technique used to map multidimensional decision support data into a relational database
  - Creates the near equivalent of a multidimensional database schema from the existing relational database
  - The schema was developed because existing relational modeling techniques, ER and normalization did not yield a database structure that served advanced data analysis requirements well
- Yield an easily implemented model for multidimensional data analysis, while still preserving the relational structures on which the operational database is built
- Has four components: facts, dimensions, attributes, and attribute hierarchies

Star Schemas

- Facts: numeric values that represent a specific business aspect or activity (sales figures).
  - The **fact table** contains facts that ate linked through their dimensions (see below)
  - **Metrics** are facts computed or derived at run time
- Dimensions: qualifying characteristics that provide additional perspectives to a fact (sales have product, location and time dimensions)
  - Dimensions are stored in a **dimension table**

Star Schema for Sales with Dimensions

FIGURE 12.12 SIMPLE STAR SCHEMA



Star Schemas

- Attributes: Each dimension table contains attributes often used to search, filter or classify facts
  - *Dimensions provide descriptive characteristics about the facts through their attributes*

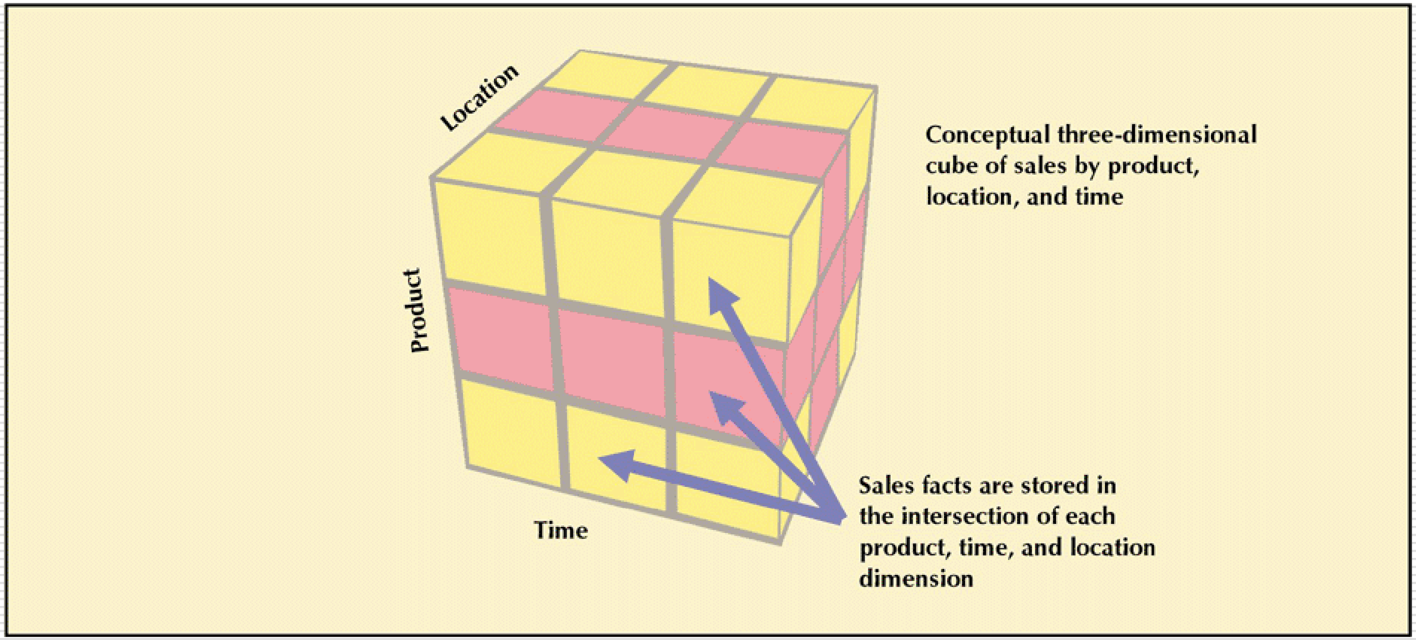
TABLE 12.9 Possible Attributes for Sales Dimensions

DIMENSION NAME	DESCRIPTION	POSSIBLE ATTRIBUTES
Location	Anything that provides a description of the location Example: Nashville, Store 101, South Region, TN, etc.	Region, state, city, store, etc.
Product	Anything that provides a description of the product sold For example, hair care product, shampoo, Natural Essence brand, 5.5 oz. bottle, blue liquid, etc.	Product type, product ID, brand, package, presentation, color, size, etc.
Time	Anything that provides a time frame for the sales fact. For example, the year 1999, the month of July, the date 07/29/1999, the time 4:46 p.m., etc.	Year, quarter, month, week, day, time of day, etc.

Star Schemas

- We can logically view the multidimensional data model as an n-dimensional cube
  - The sales data can be viewed in 3 dimensiosns – product, location and time

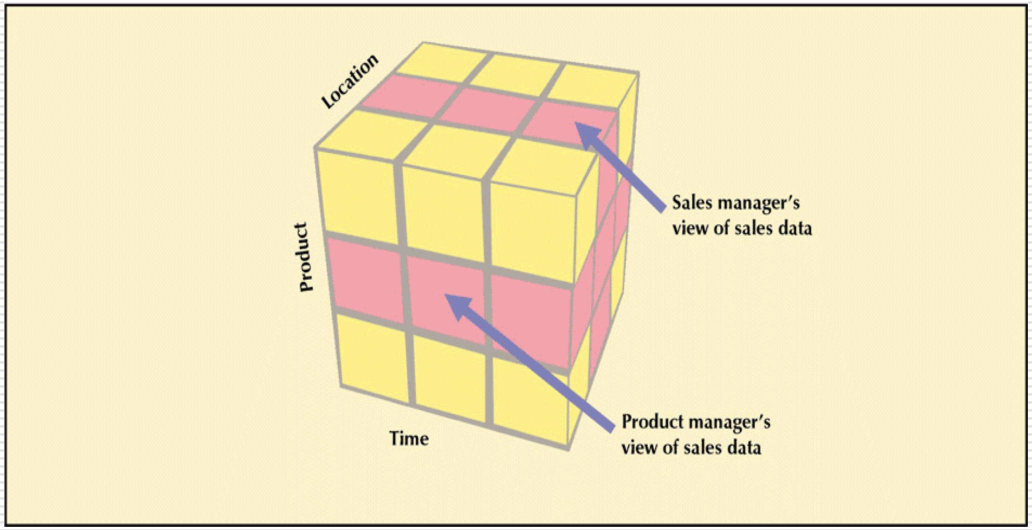
FIGURE 12.13 THREE-DIMENSIONAL VIEW OF SALES



Slice and Dice View of Sales

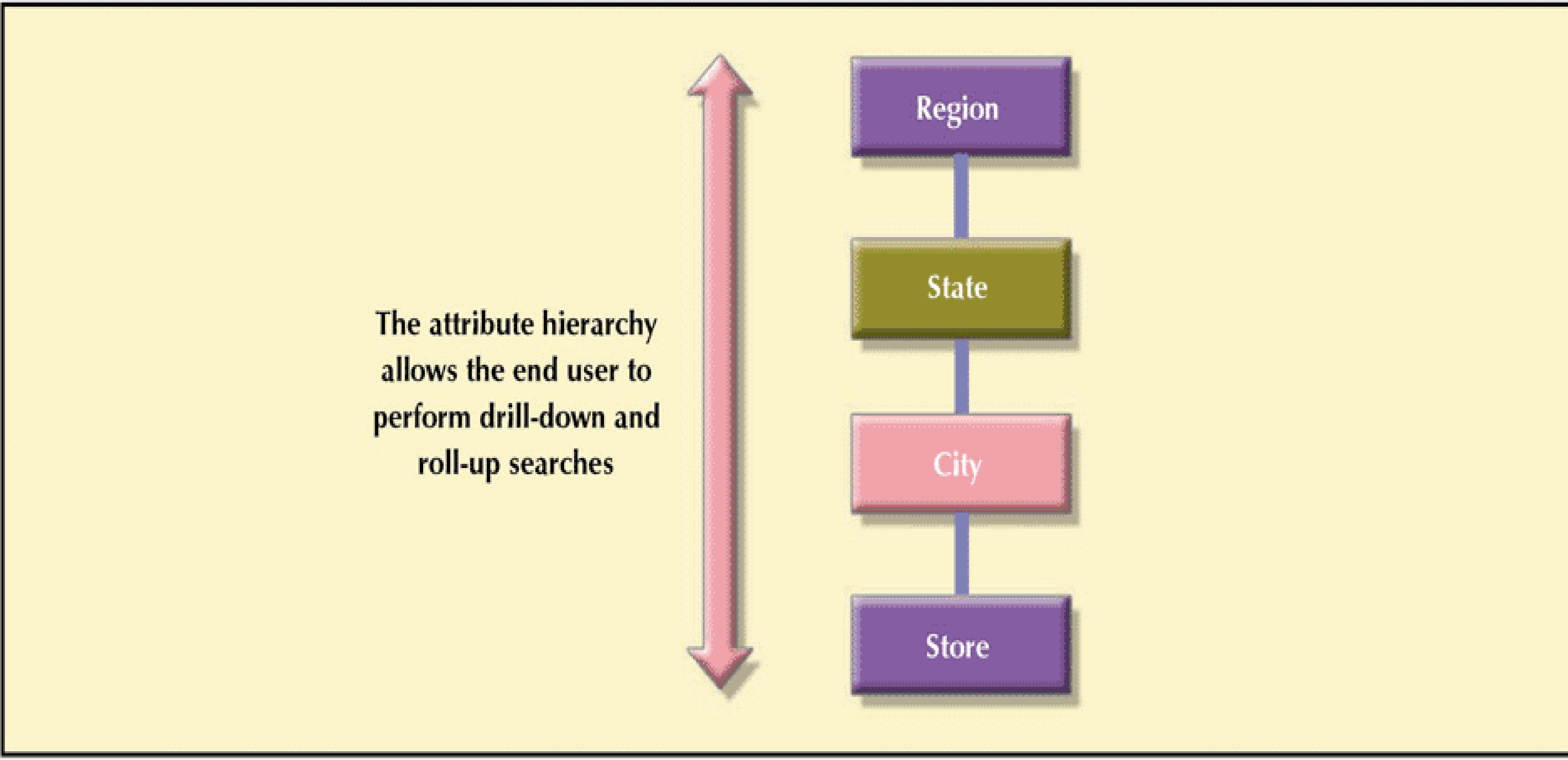
- This gives us the ability to focus on specific "slices" of the cube
  - Product manager studies the sales of a product
  - Store manager studies sales by store
  - The ability to focus on slices of a cube to perform a more detailed analysis is known as **"slice and dice"**
    - Intersecting slices produce small cubes – the dice

FIGURE 12.14 SLICE AND DICE VIEW OF SALES



Location Attribute Hierarchy

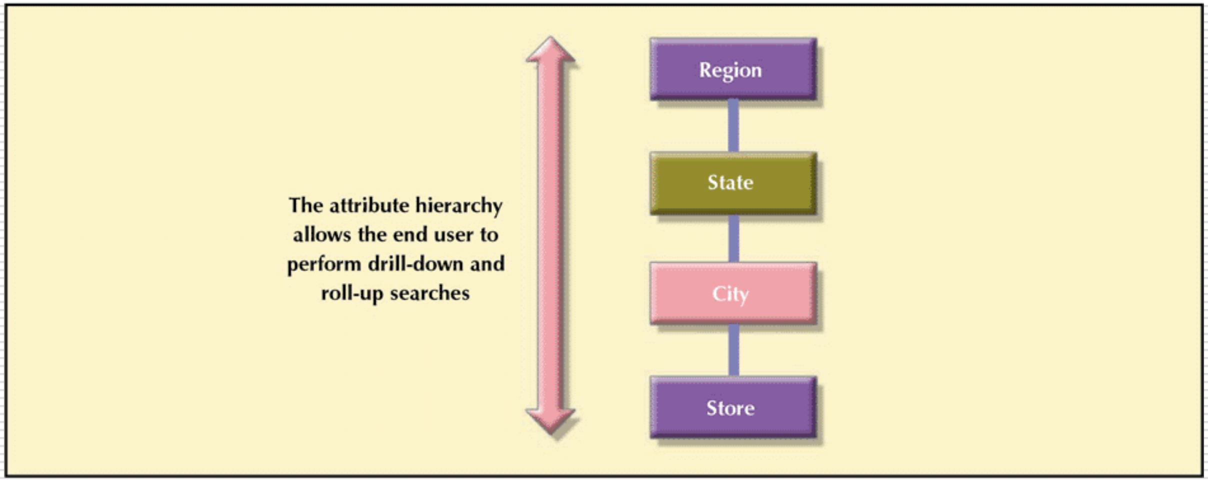
FIGURE 12.15 LOCATION ATTRIBUTE HIERARCHY



Attribute Hierarchies

- Attributes within dimensions can be ordered in a well-defined **attribute hierarchy**
- The attribute hierarchy provides a top-down data organization that can be used for
  1. Aggregation
  2. Drill-down/roll-up data analysis

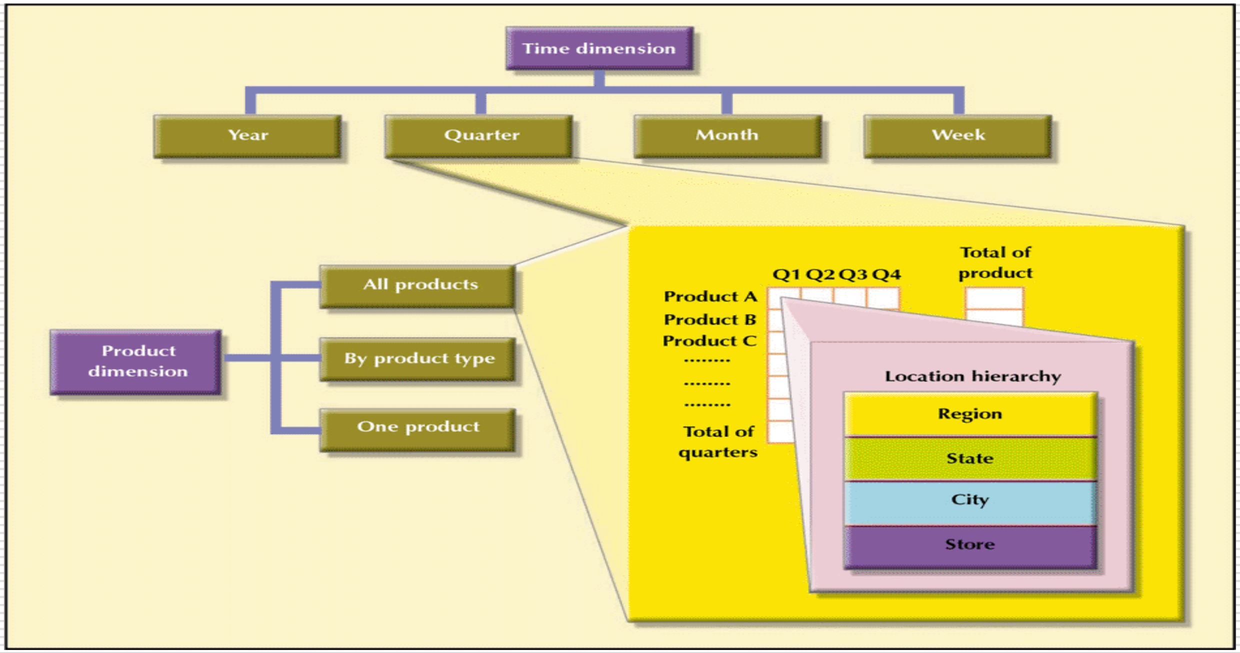
FIGURE 12.15 Location Attribute Hierarchy



Attribute Hierarchies In Multidimensional Analysis

- Product can be viewed overall, by product type or by individual product
- Time can be as detailed as a week or aggregated up to year
- Location can be as specific as store or rolled up to city then state then region

FIGURE 12.16 Attribute Hierarchies In Multidimensional Analysis

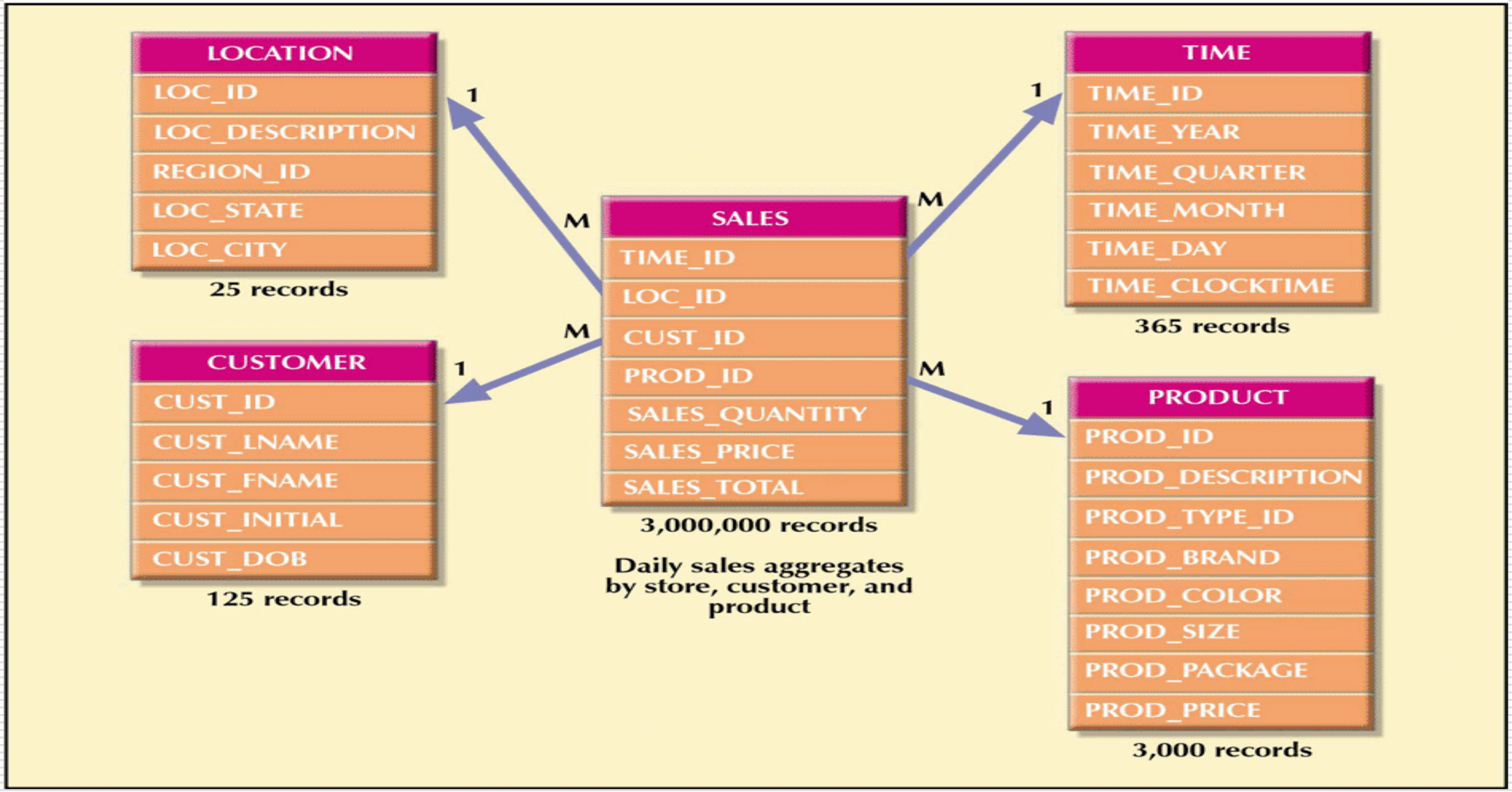


Star Schema Representation

- Facts and dimensions are normally represented by physical tables in the data warehouse database
- The fact table is related to each dimension table in a many to one relationship
  - Many fact rows are related to each dimension row – each product appears many times in the sales fact table
- Fact and dimension tables are related by foreign keys and are subject to the familiar PK/FK constraints
  - Because the fact table is related to many dimension tables, the PK of the fact table is a composite PK

Star Schema for Sales

FIGURE 12.17 STAR SCHEMA FOR SALES



Implementing a Data Warehouse

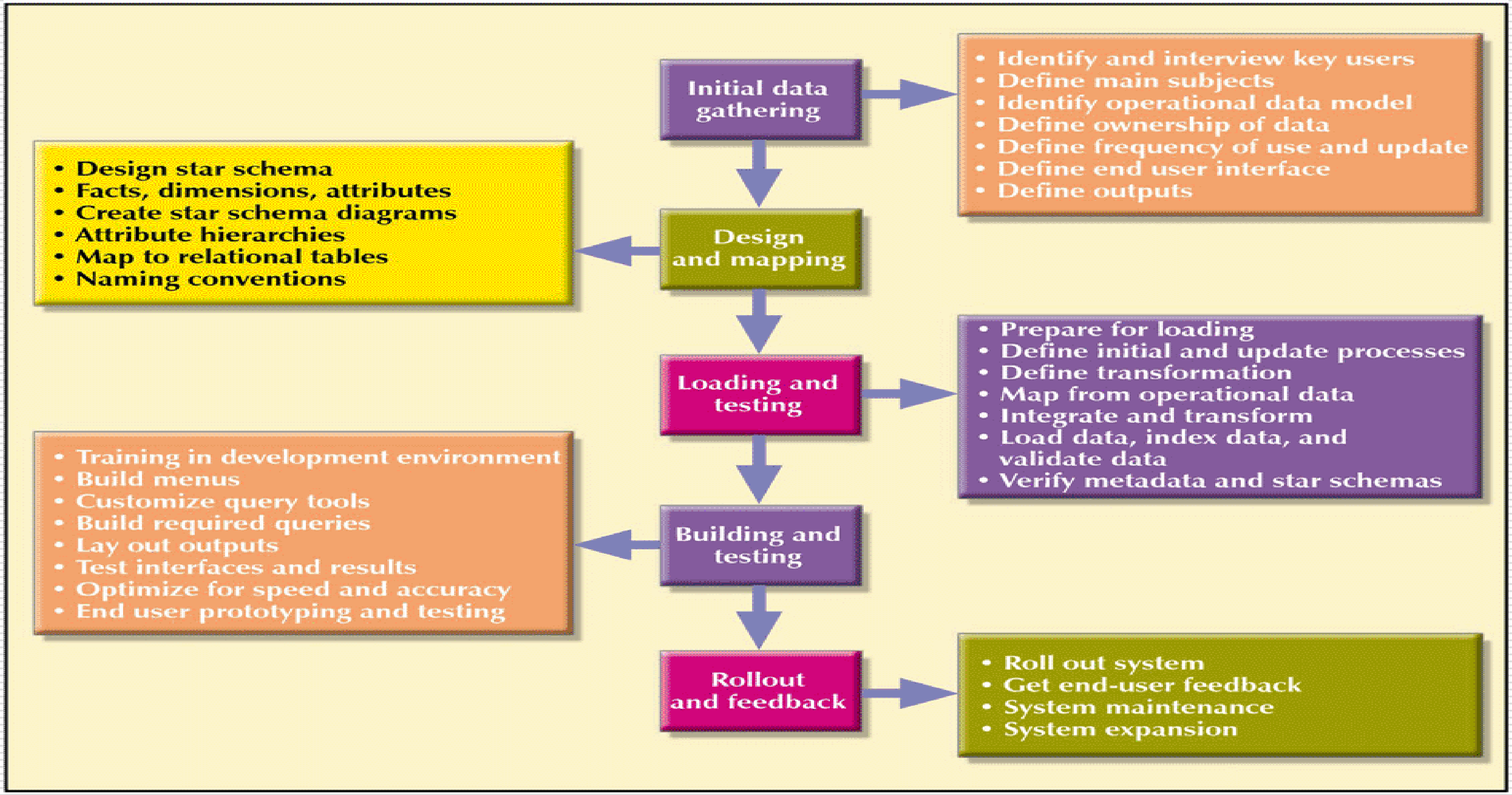
- Numerous constraints:
  - Available funding
  - Management's view of the role played by an IS department and of the extent and depth of the information requirements
  - Corporate culture
- No single formula can describe perfect data warehouse development

Factors Common to Data Warehousing

- Data warehouse is not a static database
- Dynamic framework for decision support that is always a work in progress
- Data warehouse data cross departmental lines and geographical boundaries
- Must satisfy:
  - Data integration and loading criteria
  - Data analysis capabilities with acceptable query performance
  - End-user data analysis needs
- Apply database design procedures

Data Warehouse Implementation Road Map

FIGURE 12.21 DATA WAREHOUSE IMPLEMENTATION ROAD MAP

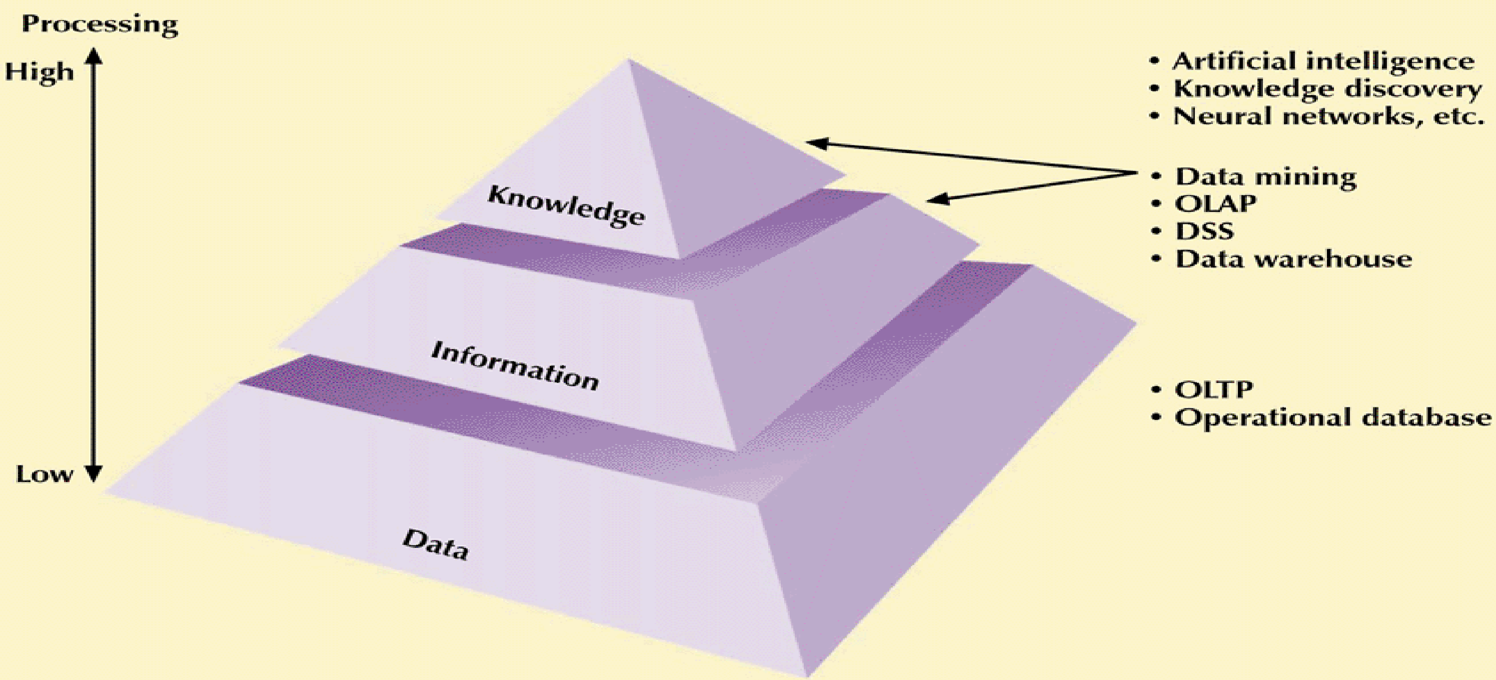


Data Mining

- Tools that:
  - Proactively and automatically search the data
  - uncover problems or opportunities hidden in data relationships
  - form computer models based on their findings, and then
  - use the models to predict business behavior
- A methodology designed to perform knowledge discovery expeditions over the database data with only minimal end-user intervention during the discovery phase

Extraction of Knowledge From Data

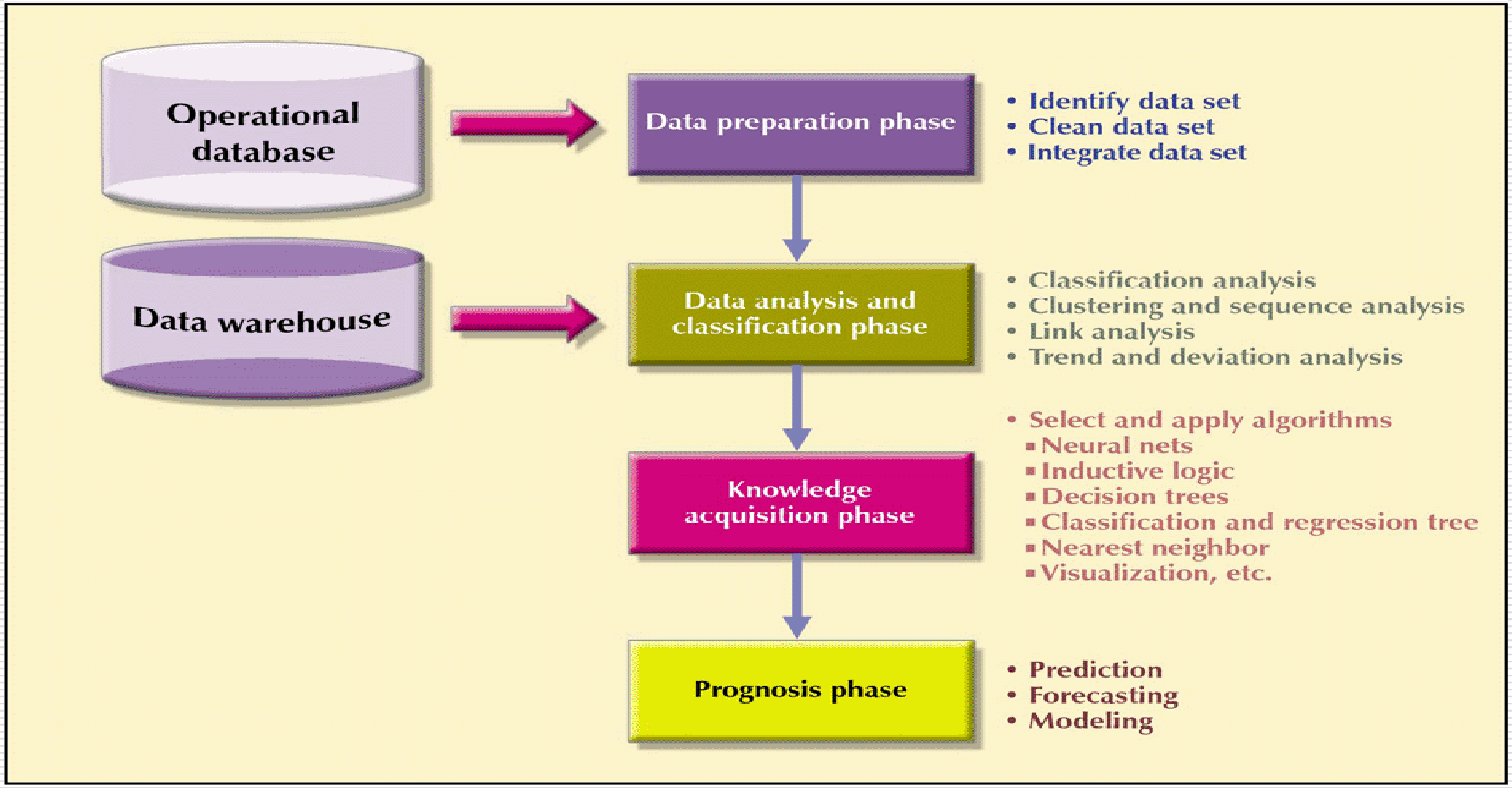
FIGURE 12.22 Extraction of Knowledge From Data



Data-mining tools use advanced techniques from knowledge discovery, artificial intelligence, and other fields to obtain "knowledge" and apply it to business needs. Knowledge is then used to make predictions of events or forecasts of values such as sales returns. Several OLAP tools have integrated at least some of these data-mining features in their products.

Data-Mining Phases

FIGURE 12.23 DATA-MINING PHASES



A Sample of Current Data Warehousing and Data-Mining Vendors

TABLE 12.10 A SAMPLE OF CURRENT DATA WAREHOUSING AND DATA-MINING VENDORS

VENDOR	PRODUCT	WEB ADDRESS
SAS Institute	SAS Data Warehouse	*www.sas.com*
IBM	DB2 Business Intelligence	*www.software.ibm.com*
Business Objects	Business Objects	*www.businessobjects.com*
Oracle	Oracle9*i*	*www.oracle.com*
Cognos	Business Intelligence Series	*www.cognos.com*
MicroStrategy	MicroStrategy 7*i*	*www.strategy.com*
Hyperion	Essbase XTD	*www.hysoft.com*